CS571 Spring 2025 – ICA I

Voice User Interface Design

Please *make a copy* of this document by clicking **File > Make a copy**. You may share and co-edit it with your fellow group members.

In this in-class activity, you will explore the concept of **Voice User Interfaces** by...

1. Reflect on Your Experiences with Chat Agents
2. Evaluate Using the Design Heuristics
3. Use Experience Prototyping for CDIS!

Areas needing your response are clearly marked with **Your Turn!** Be sure to complete all aspects of the assignment. Your Canvas submission will be a **pdf** version of this document.

<u>You may complete this in groups of 1, 2, or 3 people! :)</u> Please be sure to assign yourself and your team member(s) to a group.

1. Reflect on Your Experiences with Chat Agents

Your Turn! What have your personal experiences with chat agents been like? For *each person* in your group, please thoughtfully reflect by answering each of the following questions. There are no right or wrong answers to these questions, we just want to hear your experience!

Each person in your group should copy, paste, and respond to these questions. You can reflect on the same chat agent, even if your experiences were different!

1. What is the chat agent that you have used?

2. Is this best described as a **command-and-control** or a **conversational** agent?

3. What has your experience been like? Please comment and elaborate on any positive or negative aspects of it.

2. Evaluate Using the Design Heuristics

Having understood the basics of conversational interface design and a method to evaluate them, it's your turn to critique!

Your Turn! Having watched [the KLM Voice Assistant demo], evaluate the design using the first 8 criteria of [Speech-Based Conversational Agent Heuristics by Wei and Landay].

For each heuristic, comment (a) whether the design of the KLM agent supports or violates the heuristic and (b) your rationale.

S1. Give agent persona through language, sounds and other styles.
Your response goes here!

S2. Make system status clear.
Your response goes here!

S3. Speak the user's language.
Your response goes here!

S4. Start and stop conversations.
Your response goes here!

S5. Pay attention to what the user said and respect their context.
Your response goes here!

S6. Use spoken language characteristics.
Your response goes here!

S7. Make the conversation a back and forth.
Your response goes here!

S8. Adapt agent style to who the user is, how they speak, and how they are feeling.
Your response goes here!

3. Use Experience Prototyping for CDIS!

The idea of "experience prototyping" goes far beyond that of promoting user *interface* design – it is essential to that of user *experience* design!

Your Turn! The UW School of Computer, Data, and Information Sciences has a vested interest in making sure that the opening of the new CDIS building is a memorable and grand one. That's why they've put your team in charge! Your goal is to design and develop a Telepresence Robot experience (both the robot interface and the remote user control interface) for remote attendees so that they can view the new building environment and socialize with in-person attendees. Luckily, you know just how important it is to prototype out the experience before it happens 🙂

Here is your context. The entirety of the new CDIS is yours for the welcome day experience. How will you use it? How will you design the telepresence robot experience for remote attendees? That's up for you to decide!




Specifically, answer the following questions…

1. What does the robot look like? How would the remote users control the robot and interact with others via the robot? Be innovative and creative in your brainstorming!

2. How would you *prototype* and *test* this experience to ensure its success?

After completing this document, please be sure to upload it as a PDF in Canvas!